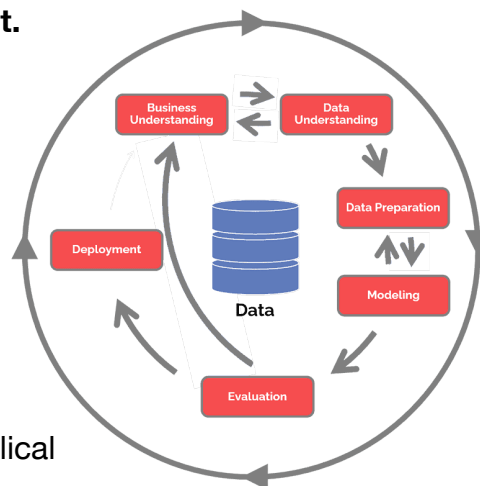


TBL: Machine Learning Workflows and Data Preprocessing

1) Consider KDD, OSEMN, and CRISP-DM frameworks for data mining. Each framework outlines a series of steps to guide the data mining process. Imagine a hypothetical scenario where a data scientist has to work on a developmental project with a tight deadline and limited computational resources. Which of the following statements is most likely to be true?

- ☐ A. The data scientist should choose the KDD framework because it has the least steps, thus saving time
- ☐ B. The data scientist should choose the OSEMN framework because it emphasizes the importance of understanding the data, which can help efficiently use computational resources
- ☐ C. The data scientist should choose the CRISP-DM framework because it is the most flexible and can be adapted to suit the constraints of the project
- ☐ D. The choice of framework does not matter because all three frameworks essentially cover the same steps but in different words and orders

2) Consider the data mining workflow presented below. From the options below, select the False statement.



- ☐ A. This framework is cyclical
- ☐ B. This framework focuses more on real-world deployment than the other frameworks covered in the lesson
- ☐ C. The modeling stage focuses on evaluating and selecting the best model for the business problem
- ☐ D. Business understanding is the initial phase in the Data Mining process, where the goal is to understand the objectives and requirements of the business

3) Consider the Dataframe shown below:

UserID	X1	X2	X3	X4	Y
0	25	Student	2	1	1
1		Student	5	2	0
2	10	Professor		3	0
3	12	Professor	4	4	1
4	15		3	5	0
5		Student	4	6	1

Having the data as is, what of the following strategies would not be adequate to handle the missing values?

- ☐ A. Using the KNN algorithm to fill X2
- ☐ B. Fill in all missing values using the corresponding column's mean
- ☐ C. Fill the missing values in X2 using the column's most frequent value
- ☐ D. All of the above are adequate approaches with the data as is

4) You have the following dataset and decide to use Average Target Encoding to make a numerical representation of variable X2. Which of the following statements is True?

UserID	X1	X2	X3	X4	Y
0	25	Student	2	1	1
1	12	Student	5	2	0
2	10	Professor	2	3	0
3	12	Professor	4	4	1
4	15	Professor	3	5	0
5	13	Student	4	6	1

- ☐ A. *Student* will be replaced with the value **0.66**, and *Professor* will be replaced with the value **0.33**
- ☐ B. Column X2 will become a binary column, where *Professor* will be represented as **1** and *Student* will be represented as **0**
- ☐ C. Since X2 only has 2 equally represented categories, both *Student* and *Professor* will be represented by the value **0.5**
- ☐ D. *Student* will be replaced with the value **3**, and *Professor* will be replaced with the value **3**

5) You are tasked with analyzing data about a set of diabetic patients staying in a hospital. The description of each column is shown in the following table:

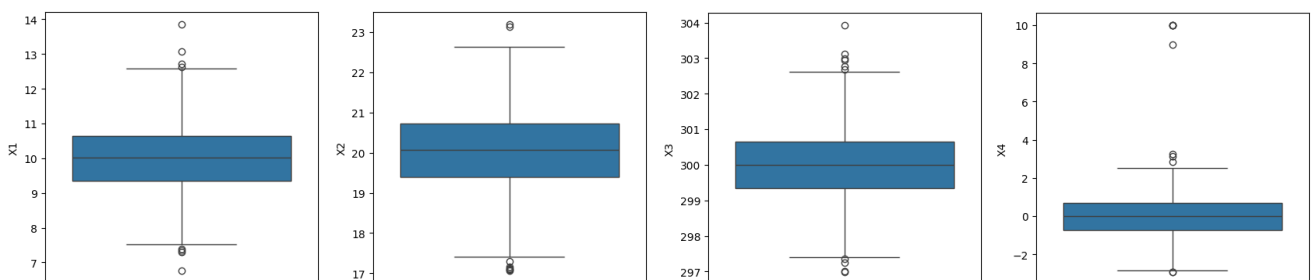
Column	Description
Pulse (bpm)	The average pulse of a patient during their stay in the hospital
Glucose Test Results	Range of the Glucose Test Results if a test was taken

As shown in the figure below, your data analysis reveals that the dataset contains missing values for patients 232 and 111. What type of missing values are you most likely in the presence of?

PatientID	Pulse (bpm)	Glucose Test Results
231	90	Normal
241	100	High
523	110	High
453	137	Normal
232	98	
111	120	

- ☐ A. Missing Completely at Random
- ☐ B. Missing at Random
- ☐ C. Missing Not at Random
- ☐ D. None of the above

6) Which outlier detection method is the most likely to be used in the figures below:



- ☐ A. Inter-Quartile Range method
- ☐ B. Z-Score method
- ☐ C. Mahalanobis Distance
- ☐ D. Cosine Similarity

7) The OSEMN framework (Obtain, Scrub, Explore, Model, iNterpret) is a common process for handling data in data science projects. Suppose you have a dataset with a significant amount of missing data and outliers. Which of the following statements is most accurate?

- ☐ A. The 'Obtain' stage would be the most appropriate for dealing with missing data and outliers
- ☐ B. The 'Scrub' stage would involve handling missing data, but not necessarily outliers
- ☐ C. The 'Explore' stage would be the best to handle missing data and outliers
- ☐ D. Both missing data and outliers should be addressed during the 'Scrub' stage

8) You use One-Hot Encoding to convert the following column into numbers. What is the minimum number of columns you would need to create to not lose any of the information available in this column?

Marital_Status
Single
Married
Divorced
Single
Widow

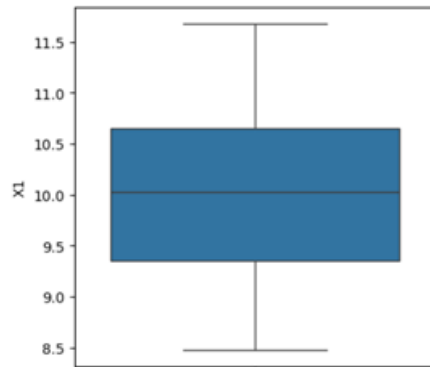
- ☐ A. 3
- ☐ B. 4
- ☐ C. 5
- ☐ D. 2

9) Feature engineering is a crucial step in the machine learning pipeline. It involves creating new features from existing ones to improve model performance. Which of the following statements about feature engineering is NOT true?

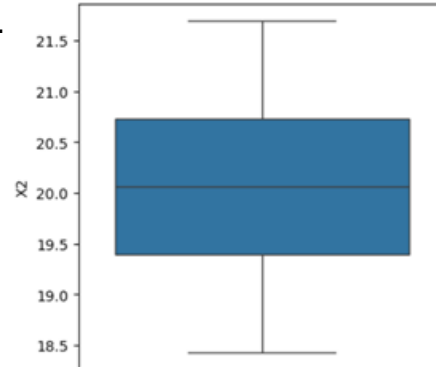
- ☐ A. Feature engineering can help to highlight the underlying patterns in the data
- ☐ B. Feature engineering can include techniques such as one-hot encoding, binning, and interaction features
- ☐ C. Feature engineering is always performed before data cleaning
- ☐ D. Domain knowledge can be very useful in feature engineering

10) Which of the following distributions exhibits outliers?

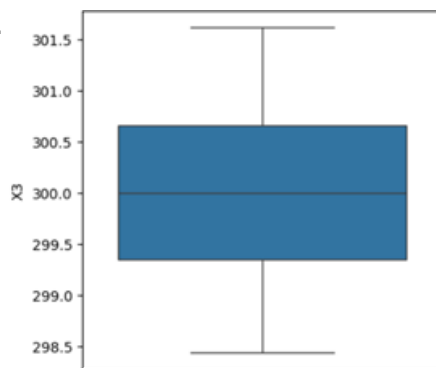
☐ A.



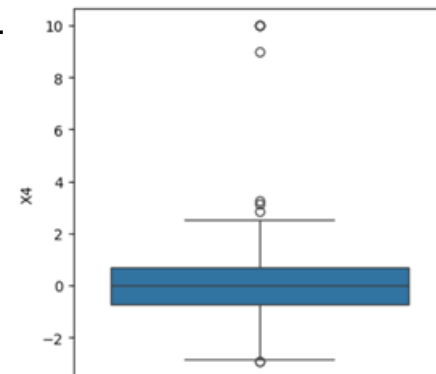
☐ B.



☐ C.



☐ D.



TBL: Model Assessment and Evaluation

1) Imagine you are working on a machine learning project to predict customer satisfaction based on survey data. The table below shows a simplified version of the dataset:

Train Data			
Index	Number of Transactions	Average Transaction Value	Customer Satisfaction Rating
1	10	\$25.00	High
2	15	\$30.00	Medium
3	5	\$100.00	Low

Test Data			
Index	Number of Transactions	Average Transaction Value	Customer Satisfaction Rating
3	5	\$100.00	Low

What is the primary issue with the split performed above?

- ☐ A. The model will likely overfit, meaning that its performance on the training data will be 3 times better on all metrics
- ☐ B. The testing set is too small, meaning that we need more observations to have a full assessment of the actual generalizability of the model
- ☐ C. We have data leakage, since, during training, the model is accessing information that is present on the test dataset
- ☐ D. The values for Number of transactions and Average transaction value are not normalize

2) Consider a dataset for predicting whether an email is spam or not. Here's a confusion matrix for a classification model. What is the precision?

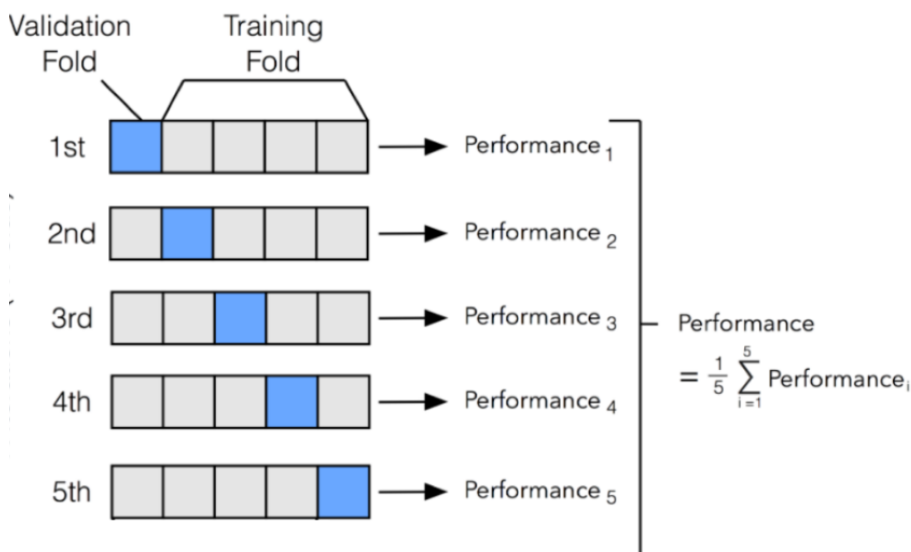
Actual\Predicted	Spam	Not Spam
Spam	50	10
Not Spam	5	80

- ☐ A. 50/55
- ☐ B. 50/60
- ☐ C. 80/95
- ☐ D. 80/90

3) Datasets are often split between training, validation and test sets. Which of the following is the correct answer?

- ☐ A. Training Set: Used to train the model, Validation Set: An additional layer of training, Testing Set: Used to challenge the model with new questions
- ☐ B. Training Set: Used to teach the model the correct answers, Validation Set: Used to confirm the model's learning, Testing Set: Used to assess the model's performance in an exam-like scenario
- ☐ C. Training Set: Used to fit the model parameters, Validation Set: Used to tune the model hyperparameters and prevent overfitting, Testing Set: Used to evaluate the model's generalization to new data
- ☐ D. Training Set: Used for initial model fitting, Validation Set: Used for final model evaluation, Testing Set: Used for ongoing model refinement

4) Which of the following options could be represented by the figure below?



- ☐ A. Leave-One-Out Cross Validation on a dataset containing 5 rows
- ☐ B. K-Fold Cross Validation when K=5
- ☐ C. Holdout Method
- ☐ D. Options A and B are correct

5) A data analyst is evaluating two regression models to predict the value of houses. The analyst has calculated the Mean Squared Error (MSE), Mean Absolute Error (MAE), and R-squared (R^2) for both models. The results are summarized in the table below:

Model	MSE	MAE	(R^2)
A	3.5	1.2	0.85
B	4.0	1.0	0.88

What can be inferred from the model's performance on these metrics?

- ☐ A. On average, Model A makes smaller mistakes in terms of absolute value, which makes it a more accurate model
- ☐ B. Model B must be better at predicting house values because a higher (R^2) value always indicates a more accurate model
- ☐ C. Model A might have fewer large errors, but Model B's predictions are more evenly distributed
- ☐ D. Model B must be preferred in all cases since lower MAE and higher (R^2) values always indicate a superior model

6) A medical diagnostics company is evaluating several machine learning models to identify patients at risk of a rare but serious disease. The primary concern is to ensure that as many actual cases of the disease are correctly identified, even if it means that some healthy patients are flagged as being at risk. The table below shows the performance metrics for four models:

Model	Accuracy	Precision	Recall	F1 Score
X	0.92	0.95	0.85	0.90
Y	0.88	0.90	0.95	0.92
Z	0.89	0.92	0.88	0.90
W	0.91	0.88	0.90	0.89

Considering the company's primary concern, what is the best model?

- ☐ A. Model X
- ☐ B. Model Y
- ☐ C. Model Z
- ☐ D. Model W

7) What is the F1 Score?

- ☐ A. The F1 score is the average of a model's precision and recall
- ☐ B. The F1 score is the harmonic mean of a model's precision and recall
- ☐ C. The F1 score is the percentage of correct positive predictions made out of all predictions
- ☐ D. The F1 score is the ratio of the model's accuracy to its error rate

8) Consider the following scenarios for a predictive classification model:

Scenario	Description
A	Identifying patients who need a specific treatment in a widespread health screening program.
B	Classifying emails in a personal inbox where most messages are legitimate.
C	Verifying entries in an evenly distributed database of biometric access records.
D	Monitoring transactions for unusual patterns within a large volume of daily financial activity.

Which of the of the following scenarios is most accurate?

- ☐ A. Scenario A
- ☐ B. Scenario B
- ☐ C. Scenario C
- ☐ D. Scenario D

9) Which of the following statements is more accurate when it comes to interpreting results from the following table? Interpret the results under perspective of the holdout method?

Data Point	Actual Outcome	Model Prediction	In Training Set?
1	Yes	Yes	Yes
2	No	No	Yes
3	Yes	No	No
4	No	Yes	No

- ☐ A. This model achieves perfect predictions on training data, meaning it will generalize well to new data
- ☐ B. The test dataset predictions perfectly match the label
- ☐ C. This model passed through an iterative cycle of training and testing that required 4 repetitions of the entire process (one for each row)
- ☐ D. The model exhibits some signs of possible overfitting

10) Which of the following statements about the R-squared (R^2) metric is false?

- ☐ A. R^2 represents the proportion of the variance for a dependent variable that's explained by an independent variable or variables in a regression model
- ☐ B. A higher R^2 value indicates that the model explains a greater proportion of the variance in the dependent variable
- ☐ C. R^2 values range from 0 to 1, where 0 means that the model does not explain any of the variability of the response data around its mean
- ☐ D. R^2 can be used to assess the correctness of the predictions made by the regression model

TBL: Feature Selection

1) To predict housing prices, a data scientist decides to use feature selection. Which of the following describes that process?

- ☐ A. Choosing a model that is most suitable for the housing market data
- ☐ B. Identifying key factors influencing price predictions, such as location and square footage
- ☐ C. Optimizing the model's hyperparameters
- ☐ D. Selecting a representative subset of observations that exemplify typical buying trends

2) What distinguishes filter, wrapper, and embedded methods when selecting features for a model?

- ☐ A. Filter methods rely on statistical measures to evaluate features independently of models, wrapper methods select the best subset of features using model performance, and embedded methods include feature selection as an integral part of the model training algorithm
- ☐ B. Filter methods use models to evaluate individual features, wrapper methods determine the importance of features based on model performance, and embedded methods assess how features interact with each other
- ☐ C. Filter methods analyze the relationships between features, wrapper methods use models to test different features sets, and embedded methods rank features according to their importance within the model
- ☐ D. Filter methods test different feature sets within a model, wrapper methods examine feature relationships, and embedded methods determine feature importance based on statistical measures

3) What are the main differences between the Pearson, Spearman and Kendall correlation indices?

- ☐ A. Pearson correlation measures the rank correlation between features, Spearman correlation measures the linear relationship between features, and Kendall correlation measures the monotonic relationship between features
- ☐ B. Pearson correlation measures the monotonic relationship between features, Spearman correlation measures the rank correlation between features, and Kendall correlation measures the linear relationship between features
- ☐ C. Pearson and Spearman correlation measure the linear relationship between features, and Kendall correlation measures the monotonic relationship between features
- ☐ D. Pearson correlation measures the linear relationship between features, Spearman correlation measures the monotonic relationship between features, and Kendall correlation measures the rank correlation between features

4) What is recursive feature elimination?

- ☐ A. A method that iteratively narrows down the feature set by eliminating the least important features based on the model's performance
- ☐ B. A technique looks at all features and assigns ranks according to their impact on the model's predictions
- ☐ C. An approach that measures the statistical correlation between different features
- ☐ D. A strategy that incrementally selects features, adding one at a time based on model accuracy

5) What are the advantages of wrapper methods?

- ☐ A. They are computationally efficient
- ☐ B. They can select the best subset of features for a given model
- ☐ C. They can account for feature interactions
- ☐ D. They are less prone to overfitting

6) What are the disadvantages of wrapper methods?

- ☐ A. They are computationally expensive
- ☐ B. They can overfit the model to the training data
- ☐ C. They are greedy methods, meaning they may miss the best combination of features
- ☐ D. All of the above

7) Suppose you are working on a project to predict housing prices based on various features, which you can observe in the following table:

Housing Prices	Square Footage	Number of Bedrooms	Location
250,000	1500	3	New York
400,000	2000	4	Chicago
350,000	1800	3	Los Angeles
500,000	2500	4	San Francisco
300,000	1600	3	Boston

Having the data as is, which of the following methods could have been used to perform feature selection across all features?

- ☐ A. Pearson correlation
- ☐ B. Spearman correlation
- ☐ C. Kendall correlation
- ☐ D. None of the above

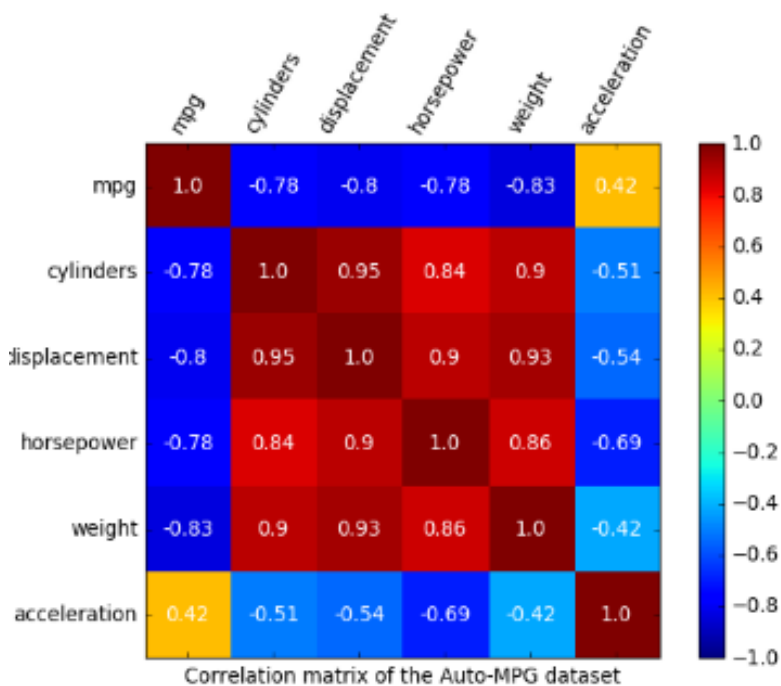
8) What is the difference between backward and forward feature selection?

- ☐ A. Backward elimination begins with the full set of features, systematically discarding the least impactful one each time, whereas forward selection starts from scratch, progressively including the most impactful feature in each step
- ☐ B. Backward elimination initially considers the most critical feature, removing the least critical one in every step, while forward selection begins with the least critical feature, adding the most critical one incrementally
- ☐ C. Backward elimination commences without any features, adding the most critical one at each juncture, whereas forward selection starts with the complete feature set, eliminating the least critical one each time
- ☐ D. Backward elimination starts with the least critical feature, excluding the most critical one at every iteration, while forward selection begins with the most critical feature, appending the least critical one progressively

9) What is stepwise selection?

- ☐ A. A filter method that evaluates the correlation between features
- ☐ B. An embedded method that selects subsets of features and evaluates their performance with a model
- ☐ C. A backward feature selection method
- ☐ D. A combination of forward and backward feature selection methods

10) The following image represents the Pearson correlation between dataset variables. According to this method, which variable is the most important predictor of the target variable (acceleration)?



- ☐ A. mpg
- ☐ B. Horsepower
- ☐ C. Acceleration
- ☐ D. Weight

TBL: Bayesian Learning Systems and Instance Based Classifiers

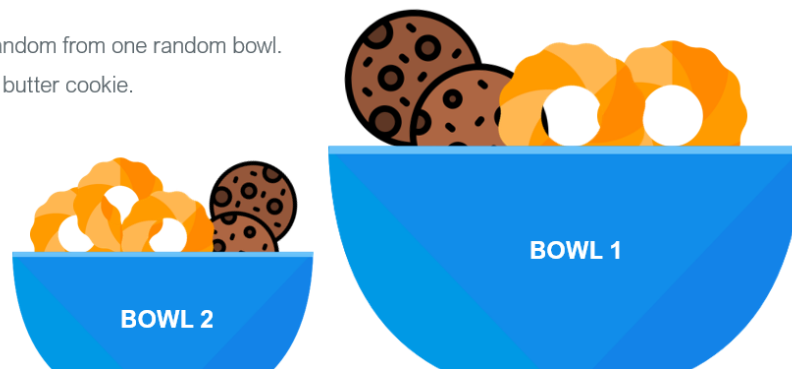
1) Consider the scenario below:

Suppose we have two bowls of cookies:

- Bowl 1 has 10 chocolate cookies and 30 butter cookies
- Bowl 2 has 20 chocolate cookies and 20 butter cookies

I just picked one cookie at random from one random bowl.

The cookie turns out to be a butter cookie.



Assuming that H_1 is “the cookie was taken from bowl 1” according to Bayes Theorem, what is the value of the prior $P(H_1)$?

- ☐ A. 0.75
- ☐ B. 0.5
- ☐ C. 0.25
- ☐ D. Can't tell from the information available

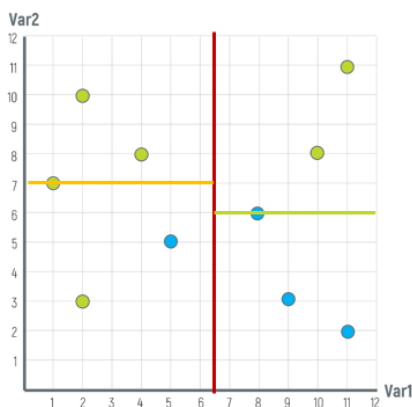
2) Which of the following statements is not an assumption made by the Naive Bayes algorithm?

- ☐ A. All features are equally important
- ☐ B. All probabilities are non zero
- ☐ C. All features are conditionally independent
- ☐ D. The statistical distribution is assumed to be known

3) In the context of classification using the Multinomial Naive Bayes algorithm, which of the following statements is incorrect?

- ☐ A. Multinomial Naive Bayes assumes that the features are conditionally independent given the class
- ☐ B. In Multinomial Naive Bayes, features are non-negative values
- ☐ C. Multinomial Naive Bayes takes into account the sequence of the features in the dataset
- ☐ D. Multinomial Naive Bayes is suitable for discrete feature values such as word counts in text classification

4) Consider the following implementation of kNN:



Var2		Var2	
8		3	
7		5	
3	→	7	→ Median = 7
5		8	
10		10	

Var2		Var2	
3		2	
11		3	
2	→	6	→ Median = 6
6		8	
8		11	

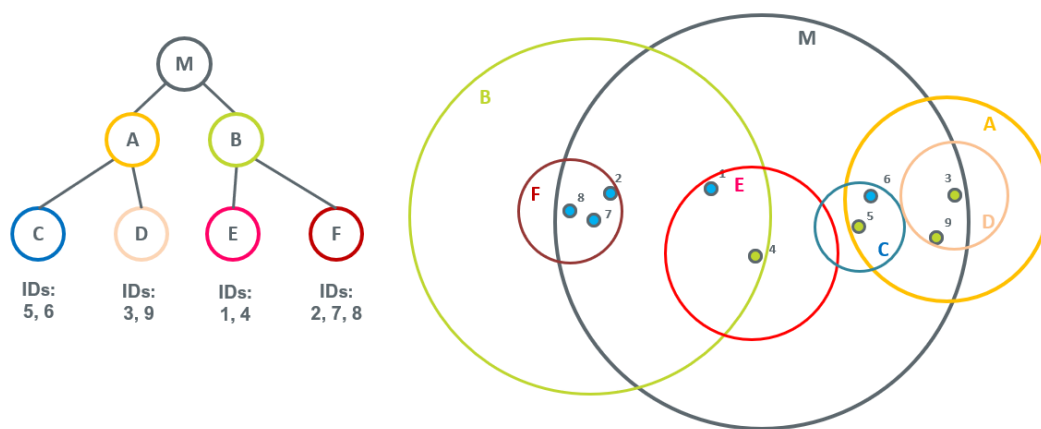
Which search method is being built here?

- ☐ A. Brute Force search
- ☐ B. Random Search
- ☐ C. KDTree
- ☐ D. Ball Tree

5) Which of the following statements best describes the difference between the ZeroR and OneR algorithms?

- ☐ A. ZeroR and OneR both consider all features in the dataset to make predictions
- ☐ B. ZeroR considers one feature to make predictions, while OneR considers all features
- ☐ C. ZeroR ignores all input features and predicts the most frequent class, while OneR considers one feature to make predictions
- ☐ D. ZeroR and OneR both ignore all input features and predict the most frequent class

6) Consider the following Trained kNN with a Ball Tree Algorithm:



Assuming a new test observation falls within circle F, why is this algorithm faster at returning the nearest neighbors in comparison to Brute Force?

- ☐ A. The Ball Tree algorithm is faster because it pre-sorts the data points based on their distances
- ☐ B. The Ball Tree algorithm is faster because it can prune large portions of the tree when searching for nearest neighbors, avoiding the need to calculate the distance to each point
- ☐ C. The Ball Tree algorithm is faster because it processes multiple branches of the tree in parallel
- ☐ D. The Ball Tree algorithm is faster because it reduces the dimensionality of the data before processing

7) Consider the dataset below:

X1	X2	Y
1	5	0
2	6	0
2	4	1
3	7	1
3	5	0

Assuming Euclidean Distance, what is the classification outcome for the following test observation ($X1 = 2$; $X2 = 5$)?

- ☐ A. Class 0
- ☐ B. Class 1
- ☐ C. Class is determined by a majority vote of its 3 nearest neighbors
- ☐ D. Cannot be determined

8) In the context of the k-Nearest Neighbors (kNN) algorithm, which of the following statements is incorrect?

- ☐ A. kNN is a non-parametric algorithm, meaning it makes no assumptions about the underlying data distribution
- ☐ B. kNN is a memory-based algorithm, meaning it requires no training before making predictions
- ☐ C. kNN uses distances as a proxy of similarity between observations
- ☐ D. Feature scaling does not affect the performance of the kNN algorithm

9) Consider the dataset below:

Outlook	Temp	Humidity	Windy	Play Golf
Sunny	Hot	False	No	No
Sunny	Hot	True	No	No
Overcast	Hot	False	Yes	Yes
Rainy	Mild	False	Yes	Yes
Rainy	Cool	False	Yes	Yes
Rainy	Cool	True	No	No
Overcast	Cool	True	Yes	Yes
Sunny	Mild	False	No	No
Sunny	Cool	False	Yes	Yes
Rainy	Mild	False	Yes	Yes
Sunny	Mild	True	Yes	Yes
Overcast	Mild	True	Yes	Yes
Overcast	Hot	False	Yes	Yes
Rainy	Mild	High	True	No

Assuming that variable *Play Golf* is the target, which of the following remarks would be true?

- ☐ A. A ZeroR algorithm would predict every new observation as **Yes**
- ☐ B. The ZeroR predictions for a new observation would depend on the value the observation has in feature **Outlook**
- ☐ C. The ZeroR algorithm would classify the new observation according to the value of the target of its near neighbor
- ☐ D. All of the above

10) Which of the following is the correct formulation of Bayes Theorem

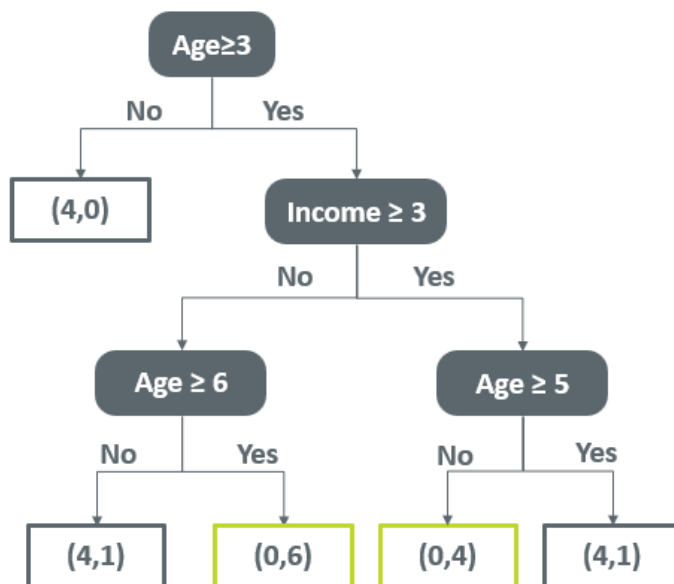
- ☐ A. $P(H|E) = \frac{P(E|H) \cdot P(H)}{P(E)}$
- ☐ B. $P(H|E) = \frac{P(H|E) \cdot P(E)}{P(H)}$
- ☐ C. $P(H|E) = \frac{P(H) \cdot P(E)}{P(E|H)}$
- ☐ D. $P(H|E) = \frac{P(E) \cdot P(H)}{P(E|H)}$

TBL: Regression and Classification Trees

1) Which of the following statements correctly describes a difference between the ID3 and C4.5 decision tree algorithms?

- ☐ A. ID3 uses entropy to measure the disorder of a set, while C4.5 uses the Gini index
- ☐ B. ID3 can handle continuous attributes, while C4.5 cannot
- ☐ C. C4.5 can handle missing values, while ID3 cannot
- ☐ D. ID3 generates a binary tree, while C4.5 generates a multi-way tree
- ☐ E. All of the above

2) As a data scientist, you train a decision tree model on a dataset and obtain the following tree:



After receiving a new observation (Age = 5, Income = 1), what would be the prediction made by this decision tree?

- ☐ A. Class 0 (Grey color)
- ☐ B. Class 1 (Green color)
- ☐ C. The prediction would be random with a chance of 50% for each class
- ☐ D. Can't tell with the information available, as I would need to know the actual value of the target variable

3) What is the main advantage of decision trees over other machine learning algorithms?

- ☐ A. Decision trees can handle both categorical and continuous data
- ☐ B. Decision trees are generally easier to interpret and explain than other models
- ☐ C. Decision trees are computationally more efficient
- ☐ D. Decision trees can easily obtain excellent performances on the training data
- ☐ E. All decision trees have built-in methods to handle missing values

4) Consider a dataset with 10 instances, 5 of which belong to Class A and 5 belong to Class B. You use this dataset to build a C4.5 decision tree. During the construction of the tree, you have the option of making one of the following 4 decision nodes, with the corresponding difference in IG relative to the current level:

Split	Information Gain
1	0.1
2	0.2
3	0.3
4	0.15

What would be the most appropriate split?

- ☐ A. Split 1
- ☐ B. Split 2
- ☐ C. Split 3
- ☐ D. Split 4

5) Consider the following dataset:

Age	#Kids	#Months as customer	...	Money Spent
18	0	1	...	5
22	1	1	...	7
23	0	3	...	6
27	2	2	...	68
34	3	2	...	72
38	2	3	...	69
43	1	2	...	77
45	0	1	...	48
48	3	2	...	42
53	2	1	...	16
57	3	3	...	18
64	1	2	...	14

You wish to create a Decision Tree that uses the last column as the target variable. Which of the following splitting criteria would you use?

- ☐ A. Mean Squared Error
- ☐ B. Information Gain
- ☐ C. Entropy
- ☐ D. None of the above

6) After creating a CART, you decide to look at the different nodes that exist in your decision Tree:

Leaf Node	Gini Index
1	0
2	0.33
3	0.66
4	1

Which of the following leaf nodes would be considered the most pure?

- ☐ A. Leaf Node 1
- ☐ B. Leaf Node 2
- ☐ C. Leaf Node 3
- ☐ D. Leaf Node 4

7) You are a data scientist working on a project that predicts whether a customer will churn based on their usage data. You decide to use CART for this task. Consider the following statements about the model you're planning to use. Which of them is correct?

- ☐ A. The model can handle both numerical and categorical variables, but it cannot handle missing values
- ☐ B. The model provides clear interpretability and can handle missing values but is prone to overfitting if not properly tuned
- ☐ C. The model can handle large datasets but struggles with high-dimensional data
- ☐ D. The model is robust to outliers but it cannot handle categorical variables

8) Which of the following statements is true about pre-pruning and/or post-pruning strategies in decision trees?

- ☐ A. Both pre-pruning and post-pruning strategies can handle missing values and categorical variables
- ☐ B. Pre-pruning requires more computational resources than post-pruning
- ☐ C. Post-pruning, which involves pruning the tree after it has been fully grown, cannot effectively reduce overfitting
- ☐ D. Pre-pruning stops the tree construction early, which can prevent overfitting, but if done excessively, it can lead to underfitting

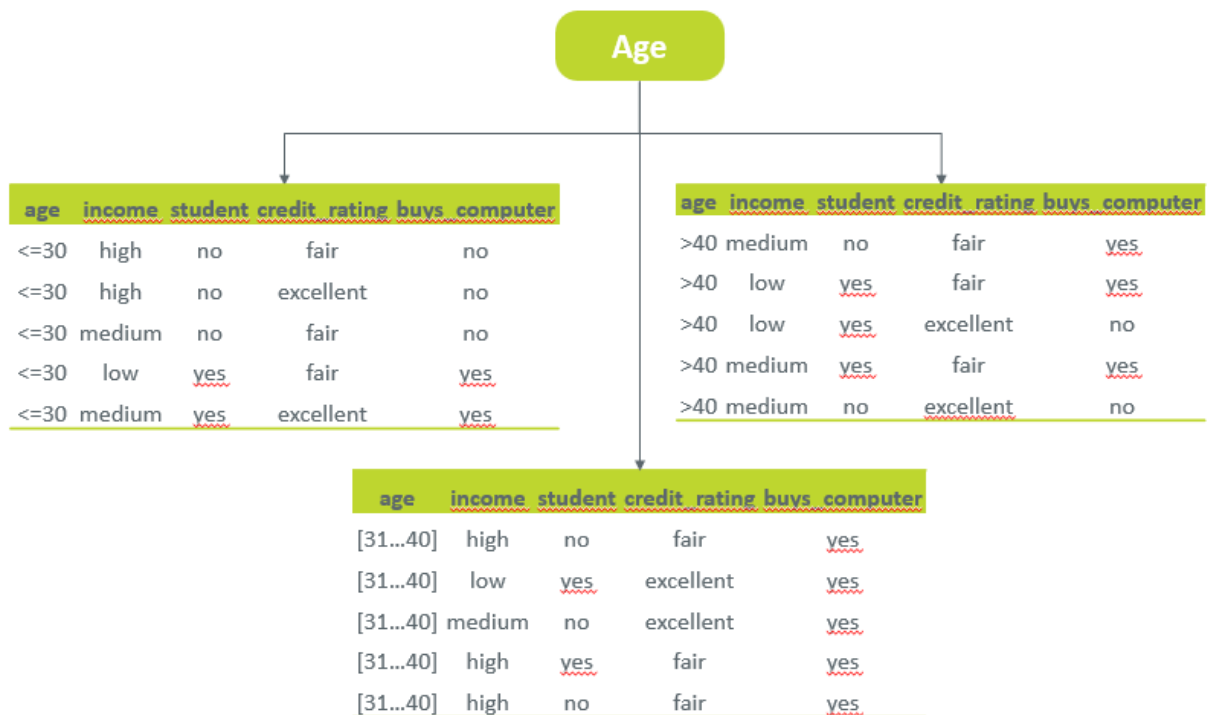
9) What does CART stand for?

- ☐ A. Conditional Analysis and Representation Trees
- ☐ B. Classification and Regression Trees
- ☐ C. Classification and Recursive Trees
- ☐ D. Categorical and Regression Trees
- ☐ E. None of the above

10) In a Dataset, you have one column called Age that can take the following unique values:

- ≤ 30
- [31...40]
- > 40

You decide to create a Classification Tree based on column Age. Which of the following Decision Tree algorithms you know **CANNOT** be used to build the tree below?



- ☐ A. ID3
- ☐ B. C4.5
- ☐ C. CART
- ☐ D. All can be used

TBL: Ensemble Classifiers

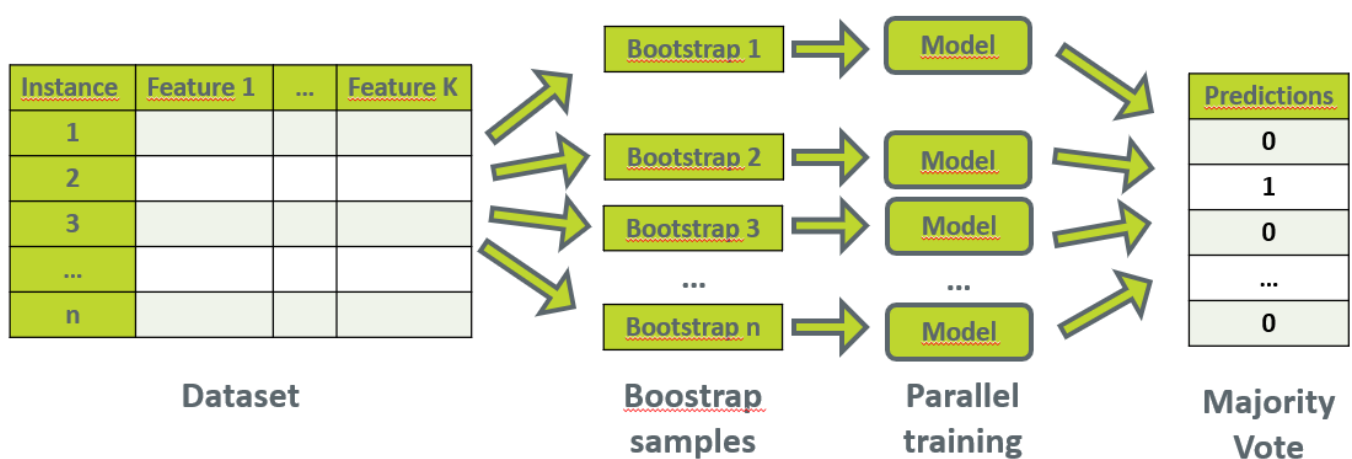
1) Which of these algorithms can never be used as the base estimator of a Bagging Model?

- ☐ A. A Logistic Regression
- ☐ B. A K-Nearest Neighbors classifier
- ☐ C. A Multi-Layer Perceptron
- ☐ D. A Decision Tree
- ☐ E. None of the above

2) Which of the following best describes the concept of ensemble learning in machine learning?

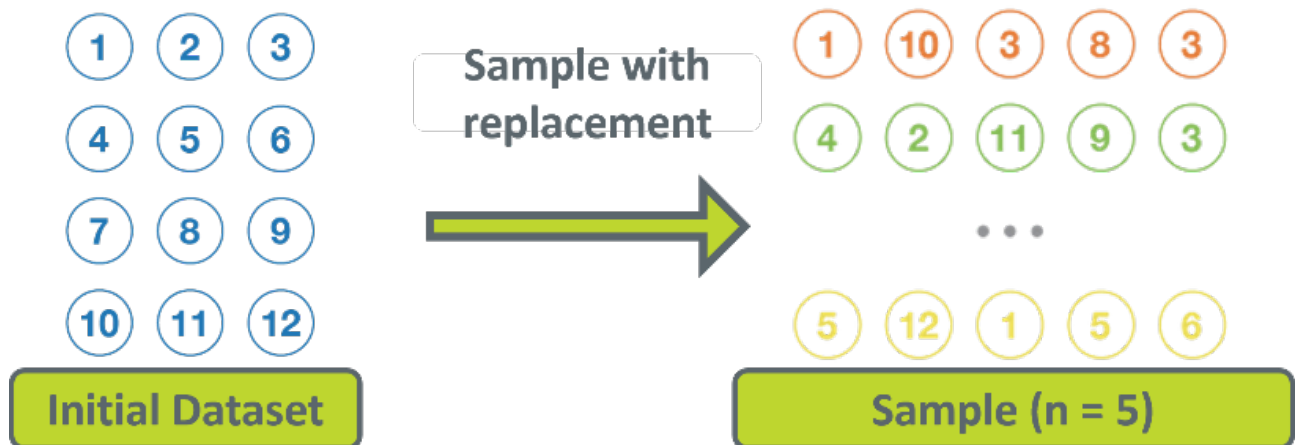
- ☐ A. An ensemble combines multiple models, while a single model uses only one algorithm
- ☐ B. An ensemble uses a separate model for each feature in the dataset
- ☐ C. An ensemble is a deep-learning model
- ☐ D. None of the above

3) What type of model ensemble is represented in the figure below?



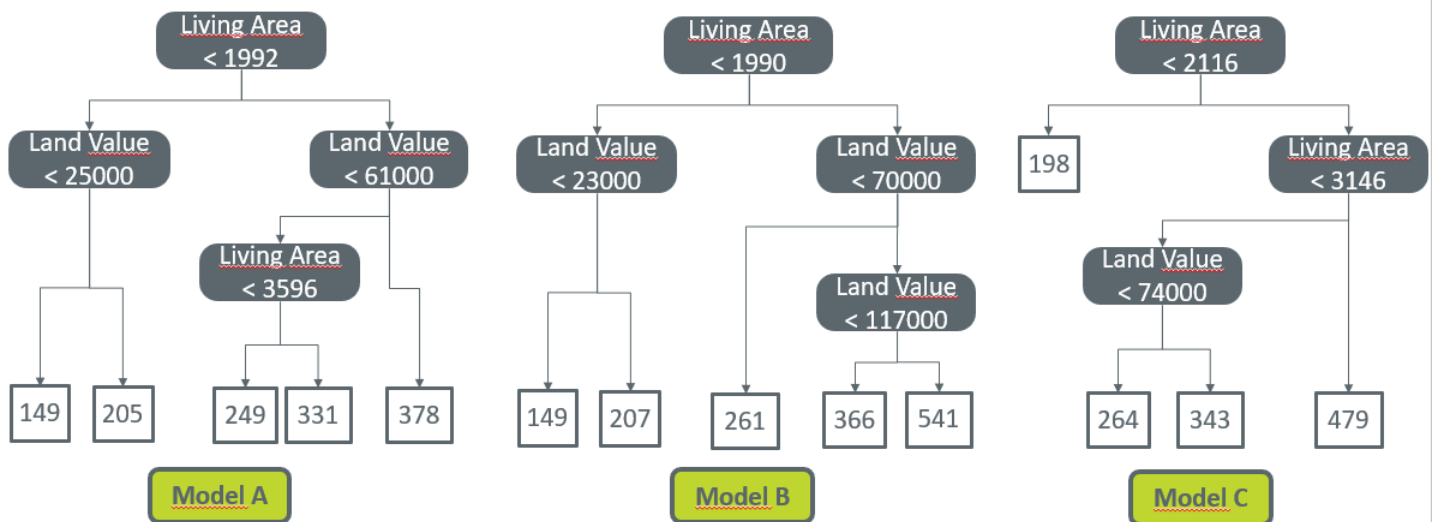
- ☐ A. A Bagging Ensemble
- ☐ B. A Boosting Ensemble
- ☐ C. A Stacking Ensemble
- ☐ D. This is not an ensemble model

4) Consider the situation below. In what type of ensembles would you expect to see this phenomenon:



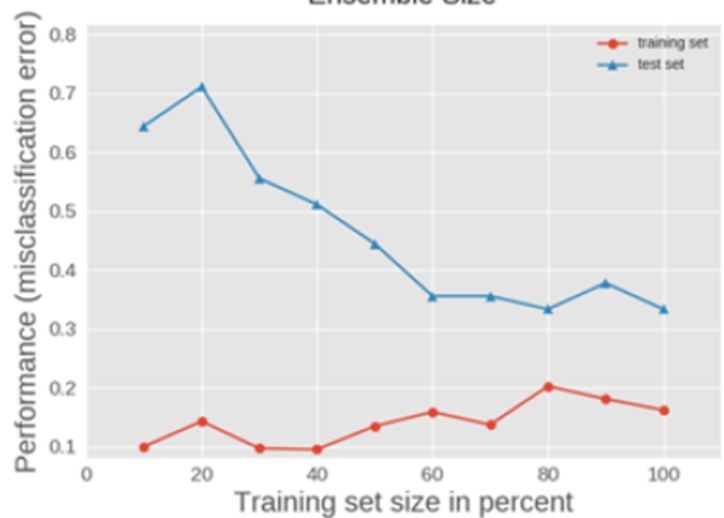
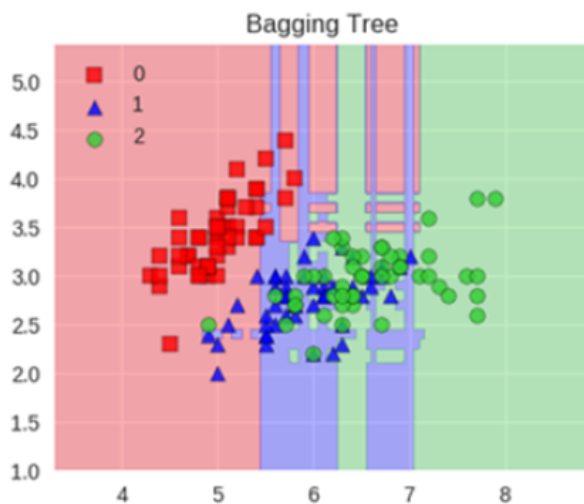
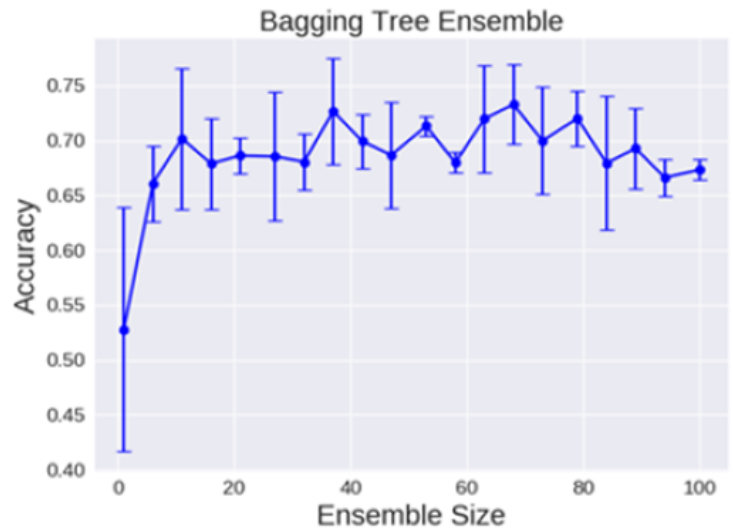
- ☐ A. Bagging Ensembles ☐ B. Boosting Ensembles
- ☐ C. Stacking Ensembles ☐ D. All Ensemble method use this approach by default

5) You have a Bagging Regressor with 3 Decision Trees that predicts House Prices. What would be the prediction for a New Sample (Living Area=3000, Fireplace=1, Land Value=50k) using the Bagging Regressor below (for each DT assume left branch means yes and right means no):



- ☐ A. 249,000 €
- ☐ B. 261,000 €
- ☐ C. 343, 000 €
- ☐ D. 264, 000 €
- ☐ E. None of the above

6) Taking into consideration the figure shown below featuring Decision Trees and Baggings of Decision Trees, what can you deduce:



- ☐ A. Adding additional trees past the 20th significantly does not seem to significantly increase the accuracy of the model
- ☐ B. Training with a smaller amount of samples seems to achieve better results on the test data
- ☐ C. These trees are solving a regression problem
- ☐ D. All of the above

7) Increasing the number of decision trees in a trained random forest model tends to:

- ☐ A. Increase computational efficiency
- ☐ B. Underfit the training data
- ☐ C. Reduce generalization error
- ☐ D. Increase explainability

8) In Random Forests:

- ☐ A. The nodes of each tree are selected randomly
- ☐ B. The number of trees is defined by the number of features
- ☐ C. Only categorical features can be used
- ☐ D. None of the above

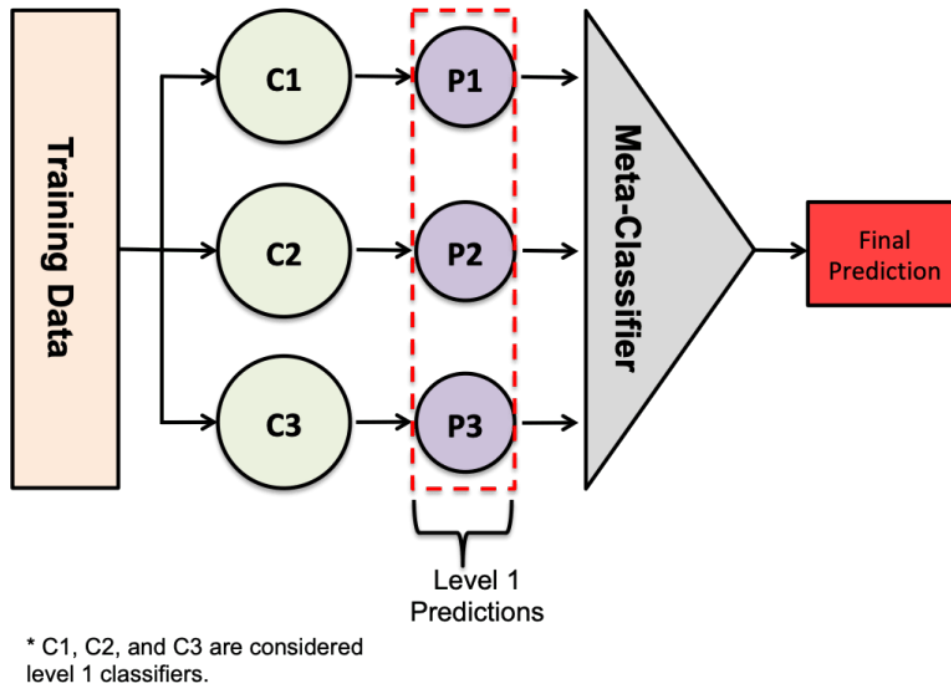
9) You decide to train an AdaBoost algorithm on a dataset. At the end of the first iteration you obtain the following weights for each observation:

age	income	student	buy	Weights
<=30	high	no	no	0.063
<=30	high	no	no	0.063
>30	high	no	yes	0.063
>30	low	no	yes	0.063
>30	low	yes	yes	0.063
>30	low	yes	no	0.250
>30	low	yes	yes	0.063
<=30	low	no	no	0.063
<=30	low	yes	yes	0.250
>30	low	yes	yes	0.063

How many samples new misclassified by the first decision stump?

- ☐ A. 0
- ☐ B. 2
- ☐ C. 8
- ☐ D. Can't tell from the information available

10) Consider the Stacking Schema below for binary classification:



Assuming C1, C2 and C3 are based learners and P1, P2 and P3 are their corresponding predictions, which of the following statements is FALSE about the meta-classifier?

- ☐ A. In a stacking architecture, the meta-classifier is a classification algorithm (e.g. a Logistic Regression)
- ☐ B. The predictions of the base learners are the *input* of the meta-classifier
- ☐ C. The meta-classifier never comes into contact with the raw data
- ☐ D. None of the above

TBL: Neural Networks I

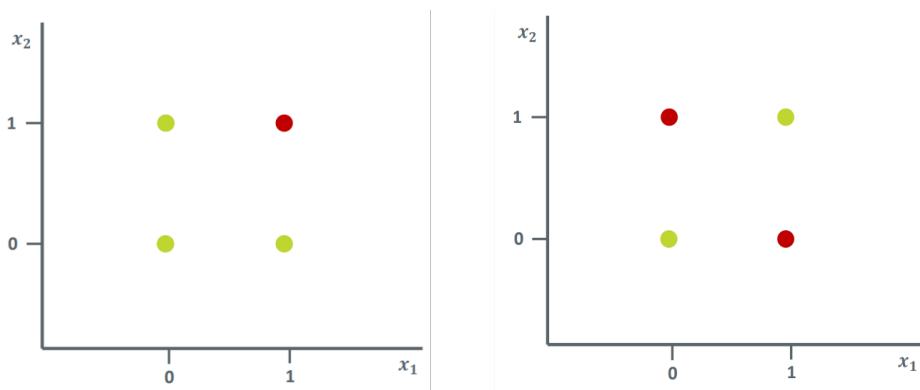
1) What is a perceptron?

- ☐ A. A type of decision tree
- ☐ B. A type of artificial neural network
- ☐ C. A type of programming language
- ☐ D. A type of database management system

2) To what biological construct is the perception compared to?

- ☐ A. A neuron
- ☐ B. A synapse
- ☐ C. A dendrite
- ☐ D. A cell body

3) Which of these classification problems (separate red dots from green dots) cannot be solved by a Perceptron?



- ☐ A. The left one, the major drawback of the Perceptron is that it cannot solve linearly separable problems
- ☐ B. The right one, the major drawback of the Perceptron is that it can only solve linearly separable problems
- ☐ C. Trick Question: the default Perceptron easily solves both
- ☐ D. Trick Question: the Perceptron by itself can solve neither. Its power only stems from the fact that it can be layered with other perceptrons to make predictions.

4) What is the output of a perceptron?

- ☐ A. A binary value
- ☐ B. A continuous value
- ☐ C. A categorical value (assuming a multi-class problems)
- ☐ D. None of the above

5) What is the main objective of the perceptron learning algorithm?

- ☐ A. To minimize the error between predicted output and the target
- ☐ B. To maximize the accuracy of the model
- ☐ C. To reduce the complexity of the model
- ☐ D. None of the above

6) What is the role of weights in a perceptron?

- ☐ A. To determine the output of the perceptron
- ☐ B. To regulate the input to the perceptron
- ☐ C. To adjust the perceptron's response to its different inputs
- ☐ D. None of the above

7) What is a multilayer perceptron?

- ☐ A. A perceptron with multiple layers
- ☐ B. A type of model used in the field of deep learning
- ☐ C. An advanced type of perceptron
- ☐ D. All of the above

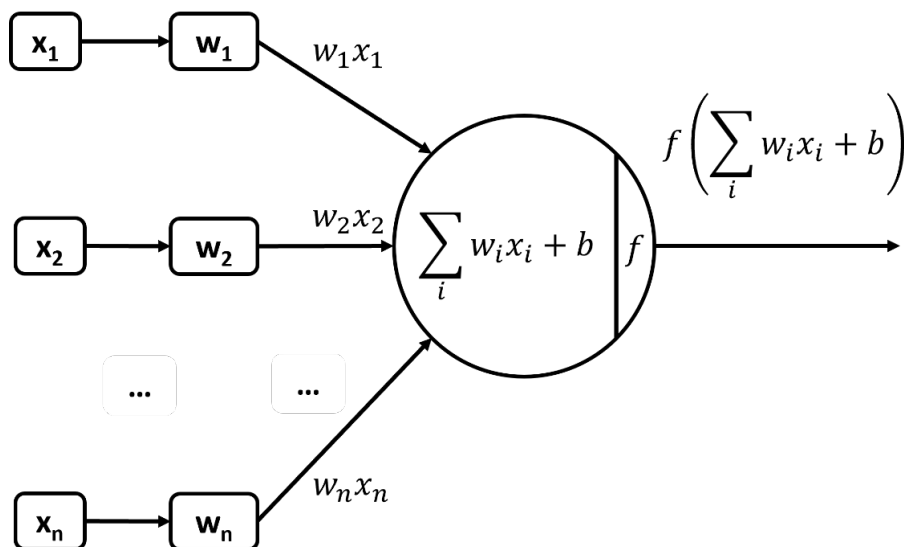
8) What is backpropagation?

- ☐ A. A type of learning algorithm used in neural networks
- ☐ B. A type of activation function
- ☐ C. A type of regularization technique
- ☐ D. A type of gradient descent

9) Which of the following is an advantage of using neural networks?

- ☐ A. They are easy to implement
- ☐ B. They can model complex non-linear relationships
- ☐ C. They are less prone to overfitting
- ☐ D. They are highly interpretable

10) What condition must be met in order for this perceptron to be used in a MLP?



- ☐ A. The activation function f needs to be linear
- ☐ B. The activation function f needs to be non-linear
- ☐ C. All weights $w_1, \dots, w_n$ need to be non-zero
- ☐ D. B and C are correct

TBL: Neural Networks II

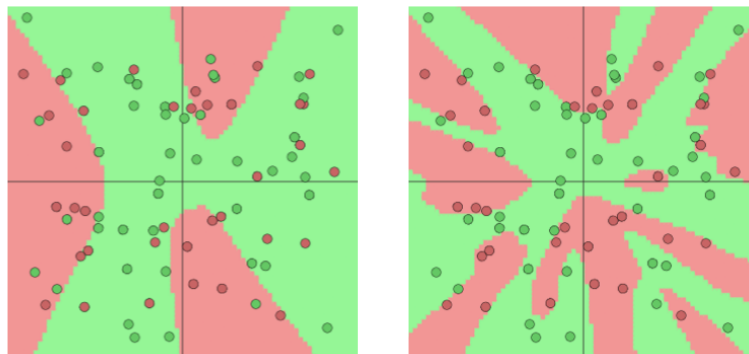
1) What is a loss function?

- ☐ A. A function that describes the possible differences between label (target) and predictions given different values of weights
- ☐ B. A non-linear function that is used to provide non-linearity to a neural network
- ☐ C. The name we give to the linear portion of the Perceptron
- ☐ D. A function that we intend to maximize by adjusting the weights of a neural network

2) Gradient Descent is...

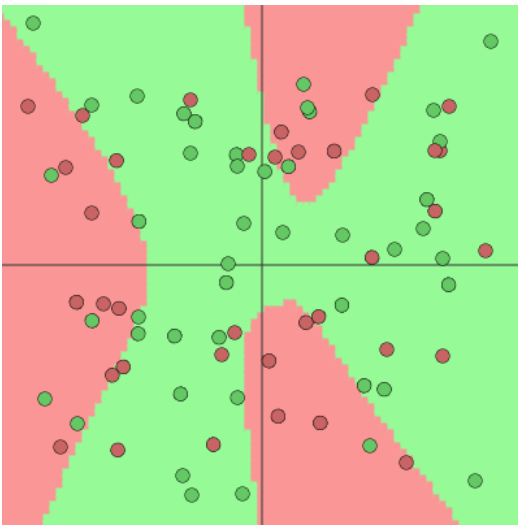
- ☐ A. An algorithm that regulates the adjustment of the weights of a neural network
- ☐ B. A different name for the forward pass of a neural network
- ☐ C. A precursor for modern-day optimizers like Stochastic Gradient Descent or Adam
- ☐ D. B and C are correct
- ☐ E. A and C are correct

3) The following decision boundaries result from 2 different neural networks with a single hidden layer. Assuming all else equal, which of the following decision boundaries is more likely to result from a network with more neurons in the hidden layer?



- ☐ A. The left one as the increased number of neurons leads to the need for more data that does not exist, thus leading to underfitting
- ☐ B. The right one as the decision boundaries are clearly tailored the training data – thus resulting in possible overfitting
- ☐ C. Trick Question: both are just as likely to be obtained by a wider network
- ☐ D. Trick Question: Everyone knows neural networks can only solve linearly separable problems

4) What is the more likely activation function used in the output layer of a neural network that results in this decision boundary?



- ☐ A. A Rectified Linear Unit (ReLU)
- ☐ B. A Sigmoid function
- ☐ C. An identity function ($y = x$)
- ☐ D. The Heaviside (or Step) function

5) What is the goal of the learning rate in Gradient Descent?

- ☐ A. To make it easier to compute the loss of a neural network
- ☐ B. To regulate by "how much" a weight should be adjusted in order to minimize Loss
- ☐ C. To regulate by "how much" a weight should be adjusted in order to maximize Loss
- ☐ D. It's just a constant value with no actual impact on neural network training
- ☐ E. None of the above

6) Adam is...

- ☐ A. An adaptive optimizer that adjusts the learning rate on-need
- ☐ B. An optimizer that keeps the learning rate constant throughout the entire training procedure of a neural network
- ☐ C. None of the above

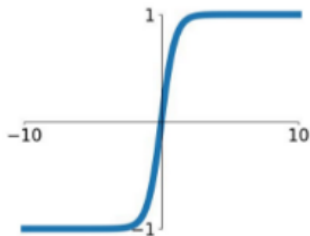
7) Complete the following sentence: Assuming that x is the input to a ReLU function, we can say that ReLU introduces non-linearity to a neural network because...

- ☐ A. Its derivative is the same for all possible values of x
- ☐ B. It has a derivative = 1 for all values $x < 0$ and a derivative = 0 for all $x > 0$
- ☐ C. It has no derivative when $x = 0$
- ☐ D. None of the above

8) Which of the following activation functions, implementable with sklearn, is not non-linear?

- ☐ A. Sigmoid
- ☐ B. Tanh
- ☐ C. ReLu
- ☐ D. Identity function

9) Which of the following activation function is described by this graph?



- ☐ A. Tanh
- ☐ B. Sigmoid
- ☐ C. ReLu
- ☐ D. Other not described in sklearn

10) Mini-batch Gradient Descent is...

- ☐ A. A variant of Gradient Descent that makes adjustments after a full run of the dataset
- ☐ B. A variant of Gradient Descent that makes adjustments after each point of the dataset
- ☐ C. A variant of Gradient Descent that makes adjustments after a certain amount of samples has been used
- ☐ D. None of the above

Answers:

	1	2	3	4	5	6	7	8	9	10
ML Workflows and Data Preprocessing	D	C	B	A	C	A	D	A	C	D
Model Assessment and Evaluation	C	A	C	D	C	B	B	C	D	D
Feature Selection	B	A	D	A	B	D	D	A	D	B
Bayesian Learning Systems and Instance-Based Classifiers	B	B	C	C	C	B	A/C	D	A	A
Regression and Classification Trees	C	A	B	C	A	A	B	D	B	C
Ensemble Classifiers	E	A	A	A	E	A	C	D	B	D
Neural Networks I	B	A	B	A	A	C	D	A	B	B
Neural Networks II	A	E	B	B	B	A	D	D	A	C